Data Visualization Pitch

THE FLOW OF HAPPINESS: A TEMPORAL TREND ANALYSIS OF GLOBAL HAPPINESS (2019–2024)

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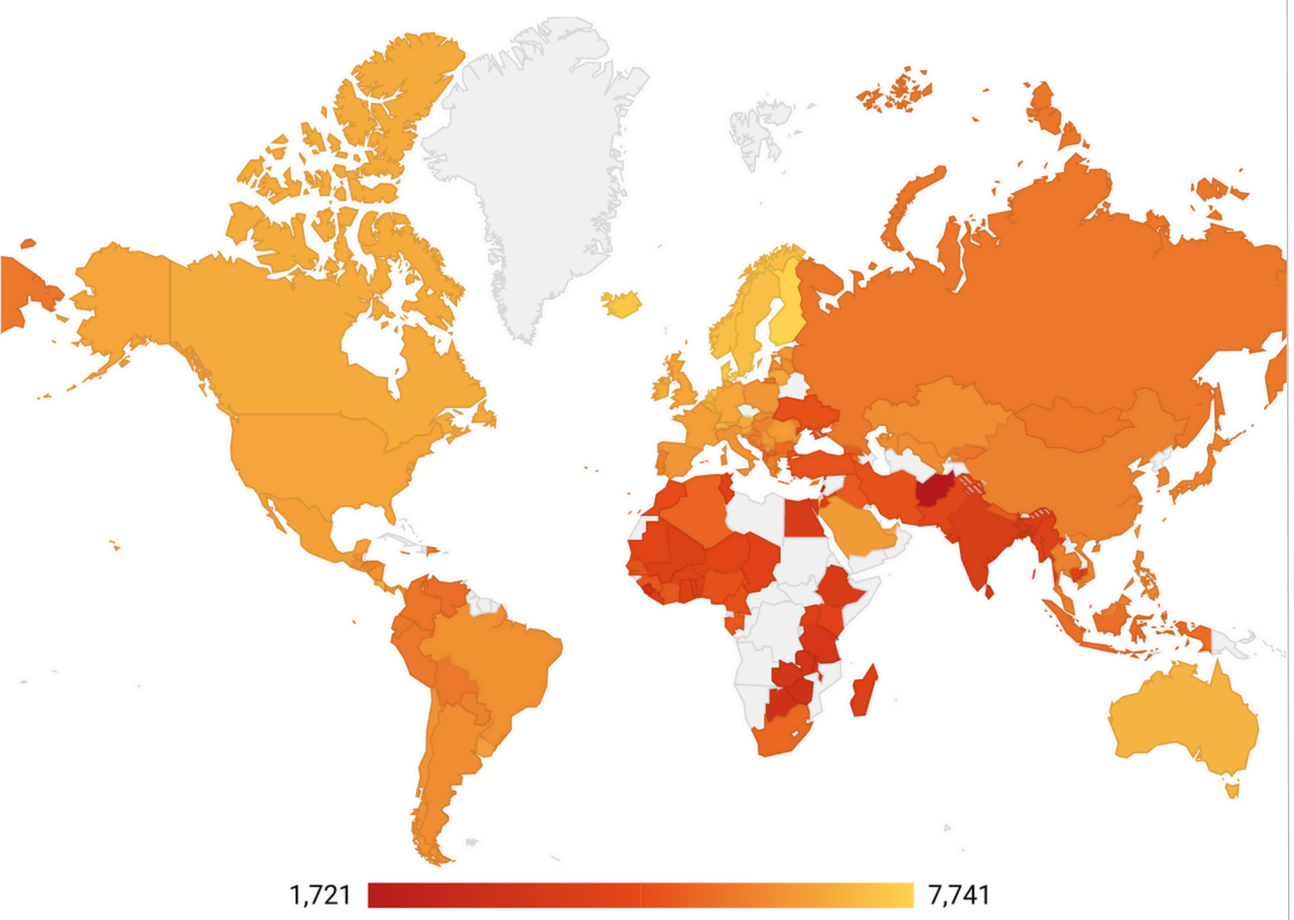


This map examines **the global happiness score** in the year **2024**, using data from *World Happiness Report* 2024. Through a continental comparative approach, the research focuses on the **whole world** to see differences with 2019.

"How and why did global happiness change from 2019 to 2024?"

RESEARCH QUESTIONS

The analysis provides an overview of **the evolution of happiness**, and the motivations for being happy in the period **2019-2024**. Some of the research questions considered are:

- How has global *happiness changed* by continent and globally from 2019 to 2024?
- Which countries analyzed have always been at the top of the *rankings* and which have always been at the bottom from 2019 to 2024?
- What are the global *motivations for being happy* and how are they related to happiness? How it changed over time?
- Did the pre-, during-, and post-*COVID* period have any influence on global happiness?

ABOUT DATA

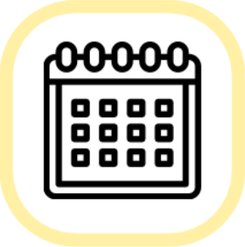
Selected Dataset: **World Happiness Report (2019-2024)**
(https://www.worldhappiness.report)

The **World Happiness Report (WHR)** collects data over the years on subjective well-being and quality of life across 157 countries. I choose to analyze the period between 2019-2024, with a mean of 146 countries per year, and the classical five global continent: *Europa, Asia, Africa, America, Oceania.*

It **contains** the annual happiness ranking, happiness score in Cantril scale 0-10, along with six socioeconomic indicators that explain life assessment: *GDP per capita* (logarithmic), *social support*, *healthy life expectancy*, *freedom of choice*, *generosity* and *perceived corruption*. It also includes confidence intervals (whiskers) and Dystopia (difference between actual and expected happiness).

It is **updated annually** and it is considered reliable. It is well-structured and relatively easy to use, requiring minimal data cleaning. But I faced **limitations**, e.g., not all countries are present in all years so globally they are not "perfectly" comparable but despite this, the difference is minimal. The other is inconsistent country names (e.g. "Turkiye" vs "Turkey") solved with pre-processing. Last but not least, analyzing only 6 years the phenomenon will not have giant changes, but they can be significant for this period.

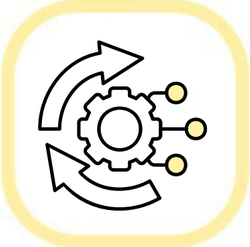
METHODOLOGY



Data collection - Rstudio
I merged six datasets, from 2019 to 2024, downloaded individually from the official website of World Happiness Report

Pre-processing & Data cleaning - Rstudio (GITHUB)

- *Missing values*: ~3.5% missingness rate, in order not to lose information the missing ones were charged with the average
- *Variable names*: renaming the socioeconomic indicators
- *Countries names:* standardization of country names



Data Transformation - Rstudio

- *Aggregated* country-level data to calculate yearly global averages
- *Grouped* countries by continent and calculated regional averages
- *Calculated* global *averages* for each of the six happiness factors across time
- *Created categorical variable* for ranking groups (Top & Bottom rank)
- *Grouped data* in pre-, during, post-covid to see the difference

METHODOLOGY

Analytical Techniques - Rstudio, Flourish, Infogram, Mapchart

- *Descriptive statistics:* calculated mean, median, and standard deviation to summarize happiness distribution; missing value patterns; identified countries consistently in Top and Bottom rankings across all years
- *Trend analysis*: examined year-over-year changes in global happiness; tracked evolution of six happiness factors over time; analyzed COVID-19 impact
- *Comparative analysis:* happiness levels across five continents; indicators of happiness over the years; contrasted Pre-COVID vs Post-COVID happiness; correlation matrix

Use of AI Tools - Rstudio, Flourish, Infogram, MapChart

- *RStudio:* for data analysis, coding, and syntax debugging with ClaudeAI (e.g. ggplot2 aesthetics, dplyr pipes)
- *Flourish, Infogram, MapChart:* for data visualization, interactive graphics, and maps
- Helped writing the comments with ChatGPT

INSIGHTS FROM THE DATA

- **Global Happiness Trends:** average happiness increased modestly (**+2%**) from 5.47 (2019) to 5.58 (2024). Standard deviation ~1.13 indicates substantial between-country variation. *Operation: Temporal aggregation*

- ***Continental Happiness Trend:*** Oceania highest (7.0) but declining **-4.1%** since 2019; Africa lowest (4.4) and stagnant; Americas biggest improver (**+3.8%**, 6.08→6.31); Asia moderate (5.3) with slight growth (+1.1%); Europe stable (**~6.5**). *Operation: Geographic aggregation, comparative analysis.*

- ***Consistent Performers:*** seven countries remained in **Top 10** throughout all six years - Finland, Denmark, Iceland, Norway, Sweden, Netherlands, Luxembourg. Three countries remained in **Bottom 10** - Botswana, Malawi, Zimbabwe. *Operation: Frequency analysis*

- **Factor Evolution Over Time: GDP** increased steadily (**+51.8%**); **Social** Support grew moderately (**+13%**, 1.16→1.31); **Freedom to make life choices** grew most (**+61.4%**, 0.46→0.75); Life expectancy declined -20.5%, likely reflecting COVID-19 mortality impact. *Operation: Factor-level temporal aggregation, percentage change*

- **COVID-19 Impact:** the pandemic devastated **key factors**: Life Expectancy collapsed -41%, Generosity plummeted -44%, Freedom declined under lockdowns. Post-COVID recovery (2023-2024) remains incomplete. *Operation: Period comparison, factor-level analysis.*

THE FALL AND RISE OF HAPPINESS

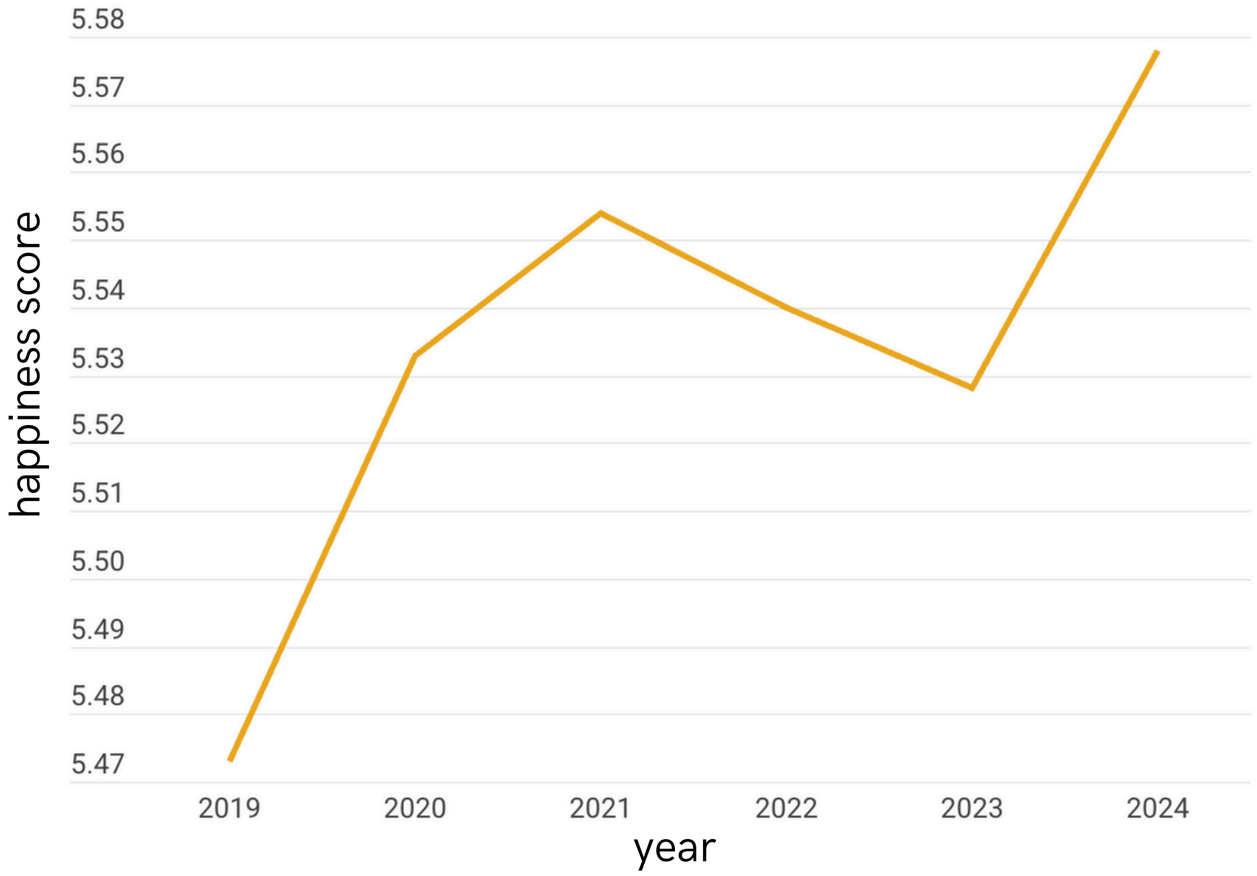
The graph shows the trend in happiness over six years, which cannot be explained in one way alone.

2019-2021
The increase likely reflects a combination of flexibility during the pandemic, reduced expectations, digital connection, and economic stimulus.

2021-2023
The decline can be attributed to: the economy has worsened due to inflation, people are less social, and fears about climate change are rising.

2024
The recovery suggests not a resolution, but an adaptation: humanity psychologically adapting to a crisis. The small overall increase masks ups and downs, demonstrating that happiness responds to both economic conditions and emotions.

Evolution of happiness score over time (2019-2024)



CREATED BY **INFOGRAM**

CONTINENTS HAVE DIFFERENT MOODS

But what's happening at the continental happiness level?
Of the five continents of the earth, the three continents below have the **most interesting trends** showing the most distinctive patterns, instead Africa and Asia that grow just a little

America

+3.8%

Continent with the best increase in happiness globally

Europa

+0.0% - stable

Continent with a high happiness score, but with a cash level over the years

Oceania

-4.1%

Continent with the highest happiness score, but with a decrease over the years

Best increase in happiness globally

Best happiness score in happiness globally

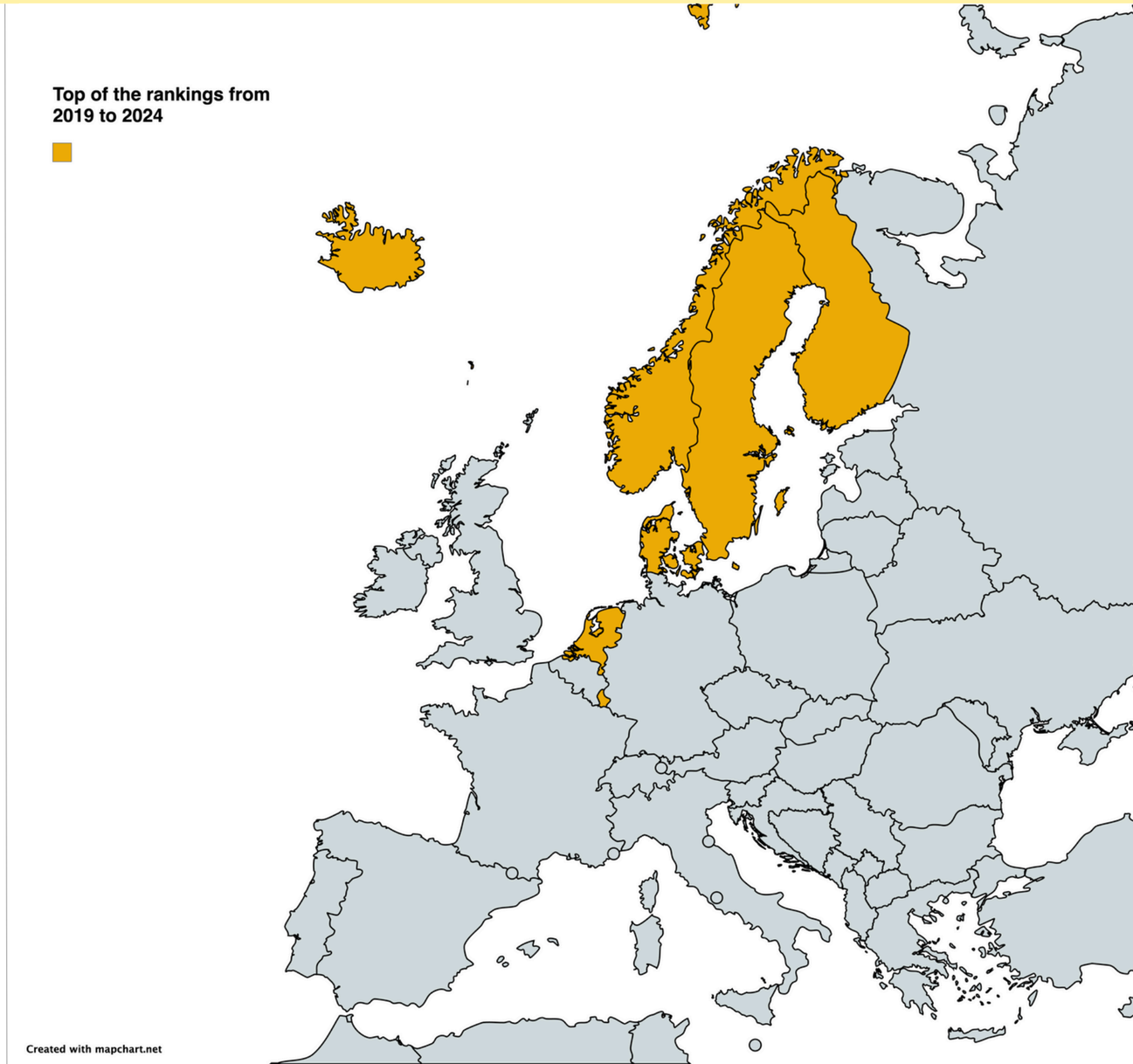
COUNTRIES CONSISTENTLY LEADERS

Europe's stability also extends to individual country performance. Seven countries have remained in the top 10 for all six years (2019-2024):

Finland, Denmark, Iceland, Norway, Sweden, the Netherlands, and Luxembourg, in gold ⟶

This remarkable coherence reveals a **"Northwest European happiness cluster"** characterized by low corruption, strong social cohesion, and democratic institutions that buffer against global shocks.

While the European continental average remained stable at around **6.5**, this map shows the concentration of countries in the North.
No Southern or Eastern European country consistently reached the top 10, **revealing persistent intra-European disparities.**



COUNTRIES CONSISTENTLY STRUGGLERS

Africa's stagnation manifests in persistent bottom rankings. Four countries remained in Bottom 10 throughout all six years (2019-2024):

Botswana, Malawi, Zimbabwe and Zambia, in red ⟶

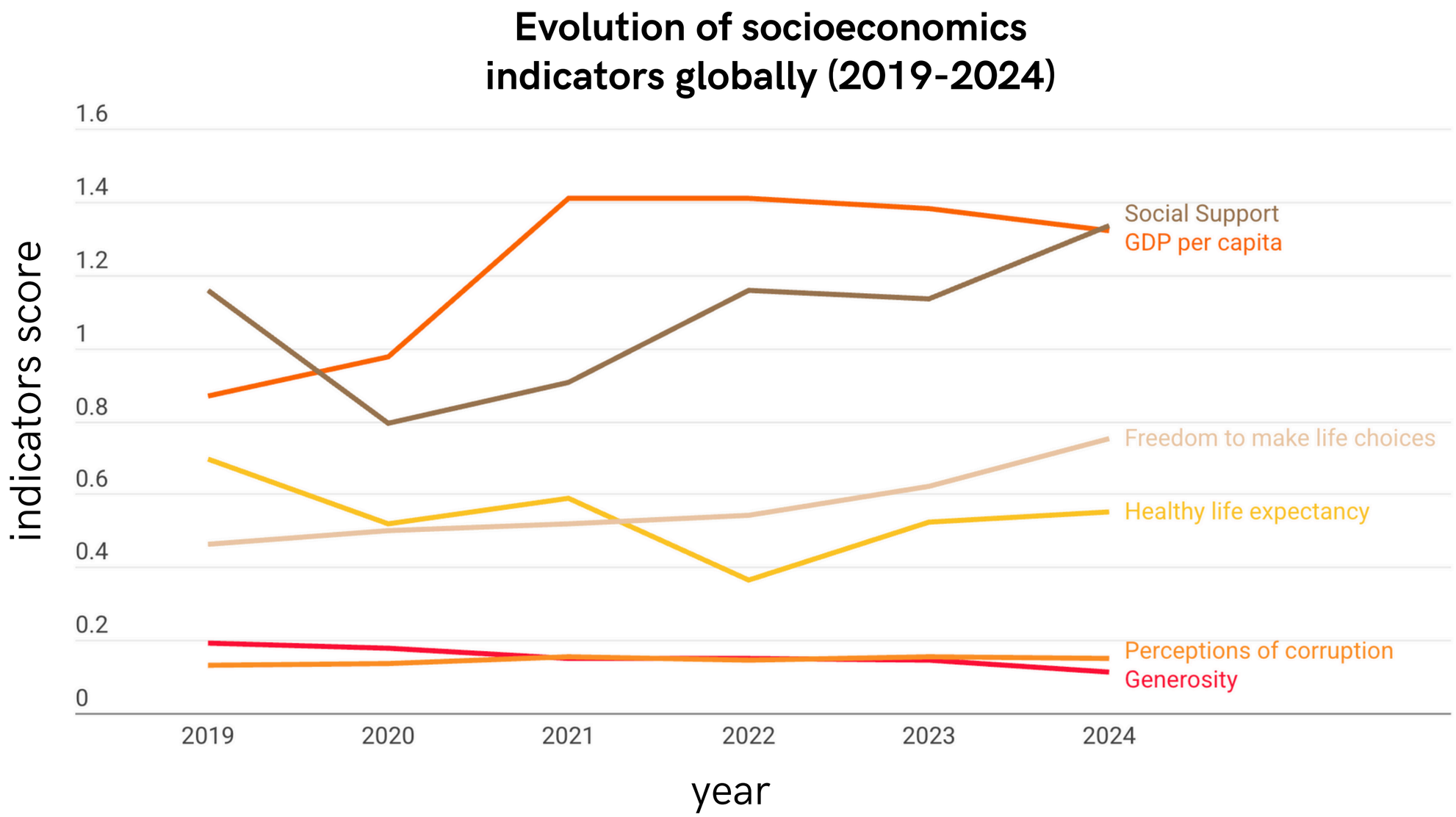
This Southern African cluster reveals a **"trap of persistent unhappiness"** where structural challenges, economic instability, health crises, governance issues, and limited opportunities, prove resistant to improvement.

Africa's suffering clusters in the **South-Central region.**

The map starkly illustrates that while Africa's continental average stagnated at **4.4**, certain nations remained trapped at the absolute bottom globally, showing no upward mobility despite six years.

Created with mapchart.net

HAPPINESS OF COUNTRIES DEPENDS ON...



CREATED BY **INFOGRAM**

HAPPINESS OF COUNTRIES DEPENDS ON…

GDP per capita

Social Support

Freedom

This graph shows a social transformation. Economic prosperity (**GDP**) and personal autonomy (**Freedom**) have increased as driving factors, while **Social** support has demonstrated resilience; **we can define them as the main drivers of happiness, but the main reasons the population is happy is that GDP.**

Generosity

What is most worrying is the steady decline in **Generosity**: this decline suggests that societies have become richer by placing importance on GDP and certainly freer, but this suggests that they are less compassionate.

The divergence between rising economic factors and social decline raises the question:
<u>are we building happiness on increasingly individualistic and materialistic foundations?</u>

CORRELATION: HAPPINESS AND ITS CAUSES

Understanding what determines happiness is crucial. Our analysis reveals the strongest predictors of happiness through **correlation analysis**, highlighting which socioeconomic factors are most closely related to life satisfaction.

Correlation

0.69

GDP per Capita

Economic prosperity strongly predicts happiness by improving access to resources, healthcare and education.

STRONGEST PREDICTOR

Correlation

0.69

Social Support

Family and community bonds provide emotional security and a sense of belonging.

STRONGEST PREDICTOR

Correlation

0.66

Life Expectancy

Health and longevity are notable predictors: longer-lived populations report higher satisfaction.

MODERATE PREDICTOR

Correlation

0.54

Freedom of Choice

Autonomy and personal agency correlate with greater life satisfaction.

MODERATE PREDICTOR

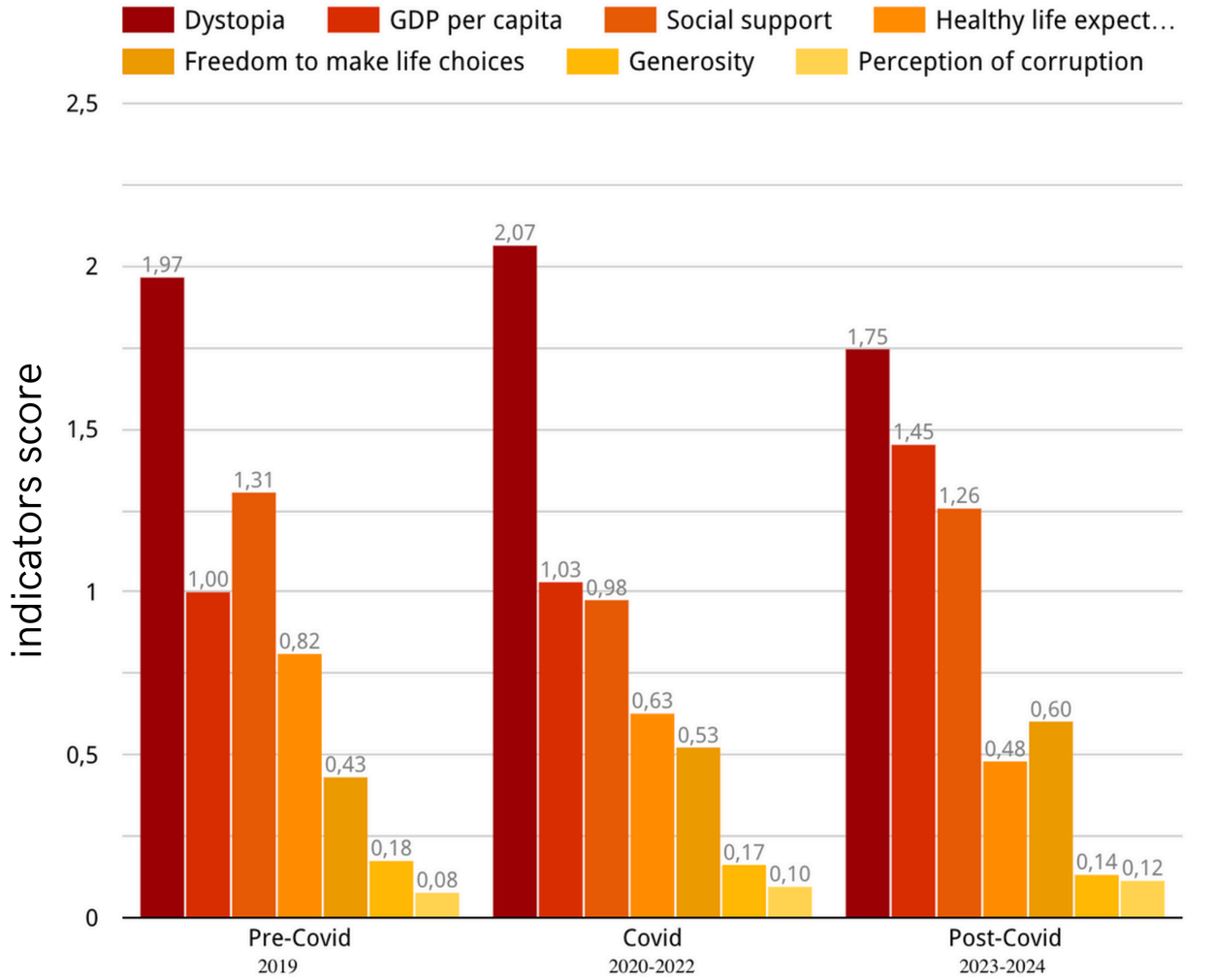
THE IMPACT OF COVID ON THE CAUSES OF HAPPINESS

In the bar plot on the right, I binned data by year, in order to obtain three timeframes: **Pre-COVID, COVID and Post-COVID.**

It would seem that the pandemic has shifted people's attention to themselves, as we can see from the **increase** in **Freedom to make life choices** and **GDP** per capita and the **decrease** of **Generosity** and **Social support**. Dystopia is an imaginary country that has the world's least-happy people. The purpose is having a benchmark against which all countries can be favorably compared. It decreases in post covid which means that the worst case scenario is now slightly better and the bottom line is less extreme.

The **heterogeneity** of factors that cause happiness shows how policies aimed at greater citizen satisfaction must take into account different needs.

How the socioeconomics indicators are changed in the three phases of COVID globally



CREATED BY FLOURISH

IN THE END

After six years of analysis, from 2019 to 2024, the data reveal a compelling **story of flexibility and progress**. Despite the unprecedented challenges of the *COVID-19 pandemic, the economy and social factors*, global happiness has seen a remarkable **recovery and growth**, heralding a good future for humanity.

Global Progress

+2%

Significant changes given the short six-year timeframe, such substantial shifts typically require decades to materialize

Continental Leaders

3 Continents

Oceania (7.0), Europe (6.5), and Americas (6.3) lead the world in happiness, establishing benchmarks for well-being and quality of life.

Key Growth Factors

+61.4%

Freedom to make life choices is the best reason of growth, instead GDP growth (+51.8%), and social support surged as well (13%)

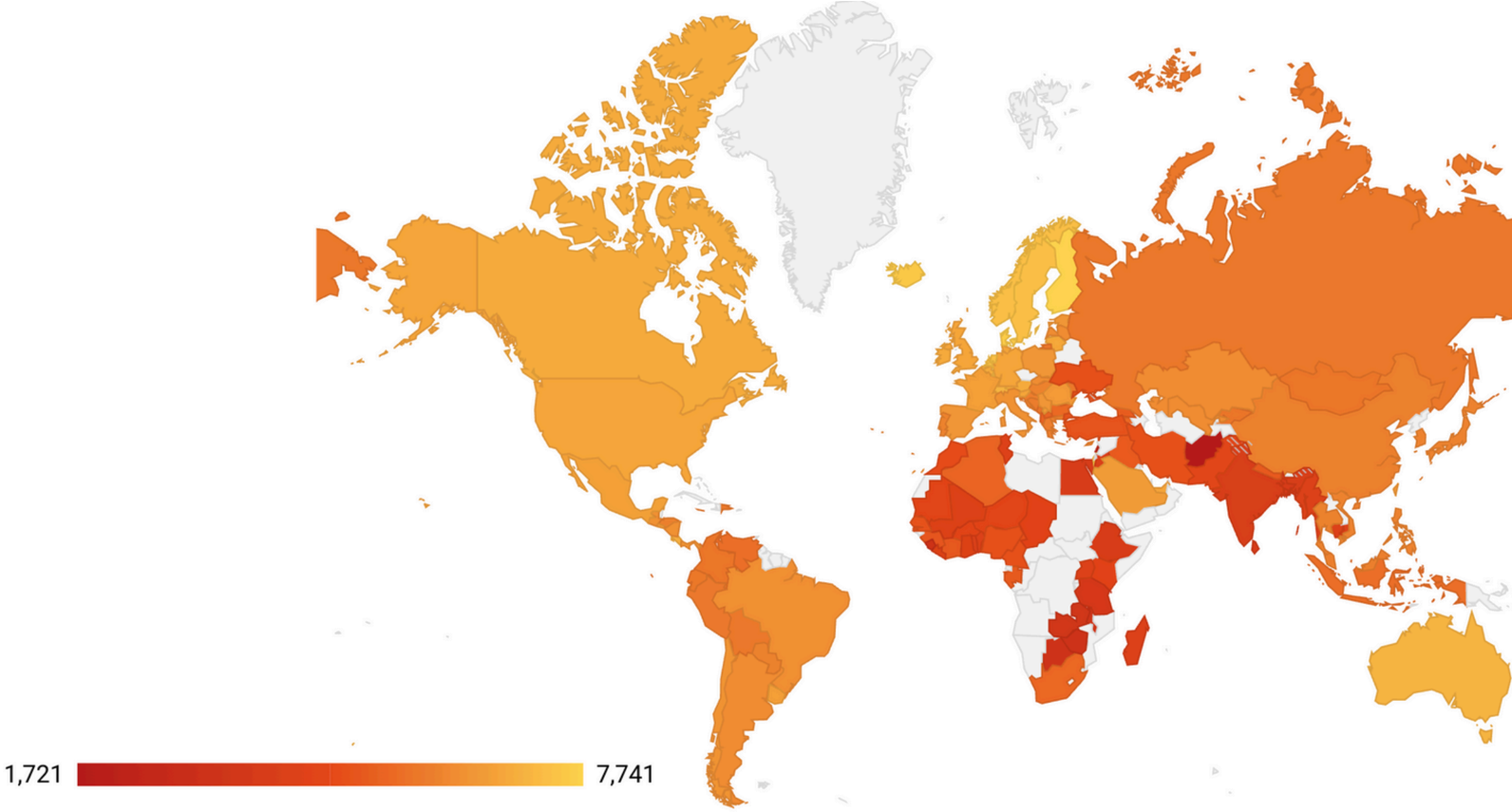
IN THE END

This **map analyzes the global happiness score for 2024**, based on data from the World Happiness Report.

Using a color-coded approach based on the 2024 happiness average, the analysis covers all regions of the world to highlight differences and trends in happiness levels.

The **goal** is to observe how happiness scores evolve across continents and identify potential regional disparities or improvements.

The continents that were found to be happiest at the end of the study were **North America, Oceania and NorthernEurope**, in accordance with the results.



LICENCE

- CC BY 4.0 World Happiness Report, Gallup World Poll, Sustainable Development Solutions Network

REPRODUCIBILITY

- Analysis is fully replicable: dataset publicly available (CC BY 4.0)
- Software environment specified (open-source libraries)
- Transformation steps explicitly described
- Replication requires downloading raw data from worldhappiness.report, applying documented cleanup operations